



Formulae & Equations

Topic Test 1 Solution

Total Marks: 30

Time: 50 minutes

1 Expand: $-2(x - 4)$ 1

$$-2x - 4(-2) = -2x + 8$$

2 Solve for x : $x + \frac{x-1}{2} = 9$ 2

$$2x + x - 1 = 18$$

$$3x - 1 = 18$$

$$3x = 19$$

$$x = \frac{19}{3}$$

3 For the equation $s = ut - \frac{1}{2}at^2$, make a the subject. 2

$$s - ut = -\frac{1}{2}at^2$$

$$\frac{s - ut}{-\frac{1}{2}t^2} = a \quad \text{OR} \quad a = \frac{2ut - 2s}{t^2}$$

4 Solve for x : $x + \frac{10+x}{3} = 18$ 2

$$x + \frac{10+x}{3} = 18$$

$$3x + 10 + x = 54$$

$$x = 11$$

5 Young's rule is used to calculate the required dosage of medicine for children aged 1 to 12 years.

$$D = \frac{yA}{y+12}$$

Where: D = Child dosage in milligrams

y = Age of child in years

A = Adult dosage in milligrams

a. What is the required dose for an 18-month-old child, if the adult dose is 72 mg? 2

$$D = \frac{1.5 \times 72}{1.5 + 12} = 8 \quad \text{Therefore 8 mg}$$

b. The recommended dose for a 3-year-old child is 5 mg. What is the adult dosage using 2

Young's rule? $\frac{3 \times A}{3 + 12} = 5$

$$\frac{3A}{15} = 5 \quad 3A = 75 \quad A = 25$$

6 Solve: $4x - \frac{2x+1}{3} = 5$. 2

$$4x - \frac{2x+1}{3} = 5$$

$$12x - 2x - 1 = 15$$

$$10x - 1 = 15$$

$$10x = 16$$

$$x = 1.6$$

7 Solve for x : $5 - 2(x + 1) = x$ 2

$$\begin{array}{r} 5 - 2x - 2 = x \\ \hline 3 = 3x \\ \hline x = 1 \end{array}$$

- 8 Ben weighs 85 kg and drinks 3 glasses of beer between 8 pm and 10:30 pm on a Saturday night. Each glass of beer is equivalent to 1.6 standard drinks. Ben's blood alcohol content (BAC) can be calculated using the following formula:

$$BAC_{male} = \frac{10N - 7.5H}{5.5M}$$

where N = number of standard drinks

H = number of hours drinking

M = male's mass, in kilograms

What is Ben's approximate BAC at 10:30 pm? (correct the answer to 3 decimal places)

$$\begin{array}{l} H = 10:30 - 8 = 2.5 \text{ hours} \\ N = 3 \times 1.6 = 4.8 \text{ standard drinks} \\ BAC_{male} = \frac{10 \times 4.8 - 7.5 \times 2.5}{5.5 \times 85} = 0.063 \text{ (3dp)} \end{array}$$

9 Given $u = \frac{\sqrt{mV}}{0.5r}$, make V the subject. 2

$$\begin{array}{l} u = \frac{\sqrt{mV}}{0.5r} \\ 0.5ur = \sqrt{mV} \\ \frac{ur}{2} = \sqrt{mV} \end{array} \quad \begin{array}{l} mV = \frac{u^2 r^2}{4} \\ V = \frac{u^2 r^2}{4m} \end{array}$$

- 10 Connor weighs 72 kilograms. He consumed 4 full-strength beers and 2 cans of pre-mixed spirits between 8:00 pm Tuesday and 12:30 am Wednesday. He then stopped drinking alcohol.

	
Full strength beer	Pre-mixed spirits
375 ml 4.8%	375 ml 5.0%
1.4 standard drinks	1.5 standard drinks

Using the formulas below, calculate the time on Wednesday when Connor's blood alcohol content (BAC) should return to zero.

$$BAC_{MALE} = \frac{10N - 7.5H}{6.8M}$$

Time for BAC to return to zero = $\frac{BAC}{0.015}$ hours

Number of standard drinks:

$$4 \times 1.4 + 2 \times 1.5 = 8.6$$

Drinking time = 4.5 hours

$$BAC = \frac{10N - 7.5H}{6.8M} = \frac{10 \times 8.6 - 7.5 \times 4.5}{6.8 \times 72} = 0.106719771$$

$$\text{Time} = \frac{BAC}{0.015} = \frac{0.106719771}{0.015} = 7.114651416 \text{ hours} \approx 7 \text{ h } 7 \text{ min}$$

Connor's BAC should be back to zero by 7:37 am Wednesday.

- 11 Juliette is driving along the highway at 100 km/h. She notices a possum on the road and applies the brakes 1.6 seconds after noticing the possum. Her braking distance is 98 metres. What is her approximate stopping distance? 2

$$\begin{aligned}
 S &= 100 \text{ km/h} = 100\,000 \div 3600 = 27.77... \text{ m/s} \\
 \text{Reaction time distance: } D &= S \times T = 27.77... \times 1.6 = 44.44... \text{ m} \\
 \therefore \text{stopping distance} &= \text{reaction time distance} + \text{braking distance} \\
 &= 44.44... + 98 = 142.44... \approx 142 \text{ m}
 \end{aligned}$$

- 12 Craig's target heart rate when exercising is calculated using the following formula: 2

$$THR = I \times (MHR - RHR) + RHR$$

where I = exercise intensity

MHR = maximum heart rate

RHR = resting heart rate

Craig wants to undertake an exercise of 0.85 intensity and achieve a target heart rate of 165.

If he has a resting heart rate of 80 beats per minute, what is his maximum heart rate?

$$\begin{aligned}
 THR &= I \times (MHR - RHR) + RHR \\
 165 &= 0.85 \times (MHR - 80) + 80 \quad \left\{ \begin{array}{l} \text{subtract 80} \\ \text{from both sides} \end{array} \right. \\
 85 &= 0.85 \times (MHR - 80) \quad \left\{ \begin{array}{l} \text{divide both sides} \\ \text{by 0.85} \end{array} \right. \\
 100 &= MHR - 80 \quad \left\{ \begin{array}{l} \text{add 80 to both sides} \end{array} \right. \\
 180 &= MHR
 \end{aligned}$$

- 13 The area of a sector is given by the formula $A = \frac{\theta}{360} \times \pi r^2$. Rearranging this formula to make r the subject. 1

$$r = \sqrt{\frac{360A}{\pi\theta}}$$

- 14 Bill has a reaction time of 1.4 s. He was driving at 72 km/h when he saw an object on the road. 2

He applied his brakes and stopped 18 m later. What was his stopping distance?

$$72 \text{ km/h} = 72\,000 \text{ m} / 3600 \text{ s} = 20 \text{ m/s}$$

$$\text{Reaction distance} = 20 (1.4) = 28 \text{ m}$$

$$\text{Stopping distance} = 28 + 18 = 46 \text{ m}$$



Formulae & Equations

Topic Test 2 Solution

Total Marks: 30

Time: 50 minutes

- 1 The volume of a pyramid has the formula $V = \frac{1}{3}Ah$, where A is the area of the base and h is the perpendicular height. Which of the following is the correct formula for A ? 1

$$3V = Ah$$
$$A = \frac{3V}{h}$$

- 2 Solve the following equations: 2
- a. $8m + 2 = 24 - 3m$

$$\begin{array}{r} 8m + 2 = 24 - 3m \\ +3m \quad +3m \\ \hline 11m + 2 = 24 \\ -2 \quad -2 \\ \hline 11m = 22 \\ \div 11 \quad \div 11 \\ \hline m = 2 \end{array}$$

b. $\frac{k-2}{4} + 5 = 7$

$$\begin{array}{r} \frac{k-2}{4} + 5 = 7 \\ -5 \quad -5 \\ \hline \frac{k-2}{4} = 2 \\ \times 4 \quad \times 4 \\ \hline k-2 = 8 \\ +2 \quad +2 \\ \hline k = 10 \end{array}$$

- 3 A formula for the dosage in mL (D) of medicine required for a child aged between 1 and 12 years (y) compared to an adult's dosage (A) is: 2

$$D = \frac{yA}{y+12}$$

$$Dy + 12D = yA$$
$$A = \frac{Dy + 12D}{y}$$

- a. Change the subject of this formula to A . 2
- b. A 6-year-old child is to take 5mL of a prescribed medicine three times per day. What is the equivalent daily adult dosage for this medicine? 2

15mL per day: $A = \frac{15 \times 6 + 12 \times 15}{6} = 45 \text{ mL/day}$

- 4 Josephine measures her mass to be 50kg. On Saturday night she had 4 standard drinks in 2.5 hours. The formula below can be used to estimate the BAC (Blood Alcohol Content) for females: 1

$$BAC_{\text{female}} = \frac{10N - 7.5H}{5.5M}$$

where N is the number of standard drinks consumed, H is the number of hours of drinking, and M is the person's weight in kilograms.

- a. Show that Josephine's BAC was less than 0.08 after her last drink. 1

$$\frac{10 \times 4 - 7.5 \times 2.5}{5.5 \times 50} = 0.07727$$

- b. 1

$$Time = \frac{BAC}{0.015}$$

The formula given above can be used to determine the number of hours required after a person stops consuming alcohol, for their BAC to reach zero. How many hours would Josephine require?

$$Time = 5.1515$$

- 5 The area of a trapezium has the formula $A = \frac{h}{2}(a + b)$. Change the subject to h . 1

$$\begin{aligned} 2A &= h(a+b) \\ h(a+b) &= 2A \\ h &= \frac{2A}{a+b} \end{aligned}$$

- 6 Christie, who was driving a van moving at 110 km/h, spotted a hazard on the road. She took 2 seconds to react before applying the brakes and finally stopped after travelling 79.3m under braking. Calculate the van's stopping distance. Give your answer correct to the nearest ten centimetres. 2

$$\begin{aligned} s &= \frac{d}{t} \\ d &= st \\ &= 110 \left(\frac{2}{60^2} \right) = 0.061 \text{ km} = 61.1 \text{ m} \\ \therefore \text{Stopping distance} &= 61.1 + 79.3 \\ &= 140.4 \text{ m} \end{aligned}$$

- 7 The formula to calculate an estimate for blood alcohol content (BAC) for males is given by 3

$$BAC_{\text{male}} = \frac{10N - 7.5H}{6.8M}$$

The number of hours it takes for BAC to reach zero after they stop drinking is given by

$$\text{Time} = \frac{BAC}{0.015}$$

Michael weighs 70kg and holds a provisional licence with a zero alcohol limit. He started drinking at 7:00 pm and consumes 9 standard drinks over 5 hours. He stopped drinking at midnight. At what time can he legally drive his car, correct to the nearest minute?

$$\begin{aligned} BAC &= \frac{10(9) - 7.5(5)}{6.8(70)} = 0.11029 \dots \\ \text{Time} &= \frac{0.11029 \dots}{0.015} = 7.35 \dots = 7\text{h } 21\text{min} \end{aligned}$$

So Michael can legally drive his car at 7:22 am*

- 8 An active woman has a basal metabolic rate (BMR) of 9 645 kJ/day. The physical activity level (PAL) for competitive cycling is 8.3. 1

$$10\,007 \text{ kJ}$$

$$\text{Energy Out} = (\text{duration of activity in hours}) \times (\text{BMR in kJ/hour}) \times \text{PAL}$$

Approximately how much energy does she burn if she competes in a 3-hour cycling race?

- 9 A patient is 18 months old. The recommended adult dose of a particular medicine is 135 mg. 2
- Use Young's formula to calculate the child's dose.

$$\begin{aligned} \text{Dosage for children 1 - 12 years} &= \frac{\text{age of child (in years)} \times \text{adult dosage}}{\text{age of child (in years)} + 12} \\ \text{Dosage for children 1 - 12 years} &= \frac{(1.5) \times 135}{(1.5) + 12} = \frac{202.5}{13.5} = 15 \text{ mg} \end{aligned}$$

- 10 a. Nathan measured the length of his back yard to be 7.2 m. What is the precision of this measurement? 1

$$0.1 \div 2 = 0.05 \text{ m}$$

- b. Calculate the percentage error for this measurement. 1

$$\begin{aligned} \text{Percentage Error} &= \frac{0.05}{7.2} \times 100 \\ &= 0.694 \dots \% \\ &= 0.7\% \quad (\text{correct to one decimal place}) \end{aligned}$$

- 11 The table shows the number of standard drinks in three types of beer available at a restaurant: full-strength, mid-strength and low-strength. The restaurant sells the beer in small and large glasses.

Number of standard drinks			
	Full-strength	Mid-strength	Low-strength
Small glass	1.1	0.8	0.6
Large glass	1.6	1.2	0.9

Geoff is 24 years old and weighs 78 kg. He

has dinner with friends at the restaurant, where he drinks three large glasses of full-strength beer and one small glass of mid-strength beer between 7:30 pm and 11:30 pm.

The following formula is used to calculate an estimate for blood alcohol content (BAC) for males.

$$BAC_{\text{male}} = \frac{10N - 7.5H}{6.8M}$$

- a. Calculate Geoff's BAC at 11:30 pm, correct to three decimal places. 2

$$7:30 \text{ pm} - 11:30 \text{ pm} = 4 \text{ hours}$$

$$BAC_{\text{male}} = \frac{10 \times (3 \times 1.6 + 0.8) - (7.5 \times 4)}{6.8 \times 78} = 0.0490 \dots \approx 0.049$$

- b. The number of hours required for Geoff to reach a BAC of 0 after he stops consuming 2

alcohol is given by the following formula. $time = \frac{BAC}{0.015}$

How long will it take for Geoff's BAC to reach zero? Give your answer in hours and minutes.

$$time = \frac{0.049}{0.015} = 3.26 \text{ hours} = 3 \text{ hours and } 16 \text{ minutes}$$

- 12 The distance, d , in kilometres that a speed camera can see when placed at a height, h , in metres above the ground is given by the formula $d = \sqrt{\frac{h}{6.5}}$.

- a. If the camera is placed 4 metres above ground, how far can it see? Give your answer correct to two decimal places. 1

$$d = \sqrt{\frac{4}{6.5}} = 0.7844 \dots \approx 0.78 \text{ km}$$

- b. Rearrange the formula $d = \sqrt{\frac{h}{6.5}}$ to make h the subject. 2

$$\begin{aligned} d &= \sqrt{\frac{h}{6.5}} \\ d^2 &= \frac{h}{6.5} \\ h &= 6.5d^2 \end{aligned}$$